

**Everyday experiments**

IKEA's labs have curated digital experiments that use cutting edge tech to rethink your home. <https://digit.in/jul20-35>

**Home of tomorrow**

This self sustaining concept home is packed with plants and has innovative eco friendly solutions. <https://digit.in/jul20-36>

Teardown:

IKEA speakers

A look inside IKEA's ENEBY and FREKVENNS speaker lineup

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**IKEA ENEBY 8"**

IKEA first entered the audio sector with this very speaker, as well as the larger 12" counterpart. However, the 8" one is also fully portable, unlike the 12" model. The 8" ENEBY can be powered by a wall plug or it can be used wirelessly using an IKEA-made 37.4 Wh Li-Ion battery pack which can be slid

into the speaker easily. The speaker comes apart after taking off 10 screws on the body. Inside you can see some connectors with locking mechanisms which need to be removed to see all the components inside. Once removed, you will be able to access a tweeter, a woofer, the motherboard, a separate breakout board for the rotary knob and line-in, and finally, a bass reflex tube. The board on the left side within the speaker provides power, while the one on the right houses the microcontroller, Bluetooth controller, and other chips.

**IKEA ENEBY Portable**

The smaller ENEBY portable speaker has a tiny, circular cover with a nifty groove that gives easy access to the speaker's battery compartment. The speaker can be powered via three standard AAA batteries or a micro-USB cable. Getting

inside this speaker is tougher than the ENEBY 8" since its screws are under a glued cover which requires some heat and prying to remove. A bass reflex tube sits within the speaker, curled around the 2.5 Watt speaker. Unfortunately, the power button, line-in, microphone, and the micro-USB port are all soldered onto the motherboard making repairs more difficult. But the battery is easily replaceable. The mainboard features a Bluetooth module, a dual-mode ROM audio SoC, a microcontroller unit, an audio amplifier, and a battery charger IC.

**FREKVENNS Portable**

The next speaker to be opened up is the tiny FREKVENNS Portable 2x4" speaker. The ultra-portable speaker comes along with a clip as well. Removing four hex screws takes apart the speaker and reveals the driver, battery pack and mainboard.

The battery can be charged via the micro-USB port which sits on the top of the speaker. The battery is connected to the mainboard with a JST plug, which can be replaced pretty easily. There is a 4 Ohm 2.5 Watt driver inside the FREKVENNS Portable. The volume encoder, as well as the line-in jack, are removable, sadly, the micro-USB port and power button are soldered onto the motherboard. Along with a few chips, the motherboard also houses an antenna in the bottom left corner.